Jess Sullivan

Senior Full-Stack & Systems Engineer · 10+ years

ML / Computer Vision · Software Systems · Full-Stack Product · Compilers & HPC · DevSecOps

Seeking on-site in Metro Boston · 617-795-6912 · jess@sulliwood.org

github.com/jesssullivan (long-standing FOSS contributor) · transscendsurvival.org · LinkedIn

Senior full stack and systems engineer with 10+ years shipping software. I love esolangs, determinism, property-based tests and learning new programming languages; my work spans TypeScript/SvelteKit, Python, Go, C++, Rust, Zig, Chapel, Haskell and Nix-first infrastructure. Current Bates work spans enterprise authentication, SAML/LTI/Shibboleth, secret management, ACME certificate automation, OpenTelemetry, GitLab AutoDevOps, OpenTofu, RKE2/Rancher, legacy modernization and 24/7 CVE response. Earlier work shipped Merlin Sound ID and its tooling/demos for fine-grained audio classification used by millions. FOSS and client work spans compilers, GIS, ML and systems tooling. Comfortable moving between ML tooling, user-facing product surfaces, HPC and large-scale infrastructure to low-level hacking.

Always building: Hey! I am *always* learning, building, reading and tinkering. Day in, day out, this is what I do. For a more up-to-date view into what I am up to whenever you are reading this document, I invite you to explore my blog at transscendsurvival.org and recent commits on github.com/Jesssullivan.

Product / Applied ML	**Systems**	**Languages**	**Build / CI / Infra**
TensorFlow, Starlark	OS internals, ABI / FFI	Python, C / C++, Rust, Zig	Bazel, Nix flakes, OpenTofu
Fine-grained classification	Ghidra / GhidraScript	Chapel, Haskell, Futhark	LLVM / Clang, LLVM IR, Justfile
ML annotation & MLOps tooling	Frida, Wireshark	TypeScript, Go, R/Rcpp, Bash	GitLab CI, Podman/OCI, RKE2
Expert in SvelteKit; WebGPU	ACME certificate automation	Objective-C, Swift	OpenTelemetry, Tempo, SOPS+age

Selected Systems, Compilers & Applied R&D Work

Applied ML:

- **Merlin Sound ID**: developed and launched a production fine-grained audio classification system used by millions; Objective-C, Swift, Python/TensorFlow training, annotation, model-evaluation and MLOps tooling (role detail in §2).
- **MushroomObserver, Visipedia**: contributed to novel community-science fine-grained ML classification research, including NSF-supported work.

Compiler & toolchain:

- **Chapel-lang (LLVM-21 backend)**: contributor to the Chapel Mason registry and downstream consumer of Chapel's LLVM-based code-generation pipeline. Author of a property-based test harness for Chapel with QuickCheck-style shrinking over distributed code paths.

Kernel & low-level security research:

- **Firmware reverse engineering**: NVMe XRAM injection recovery research and ASM2362 experiments using GhidraScript, C, Zig and inline assembly.
- **Focusrite FireWire interfaces**: ongoing reverse engineering of all interfaces and their drivers, ported to Apple silicon and real-time Rocky Linux. Private work; ask me about this project!
- **Zig de-attestation libraries**: cryptography, desktop notifications, OS keychain access and FIDO2 / WebAuthn, with C ABIs.

FOSS:

Active contributor to: Apache Solr, KeePassXC, rspamd, Chapel Mason registry, llama.cpp, Klipper, FFT.js, nixpkgs, numtide/nix-vm-test, cmux, Futhark, xCaddy, svelte-superforms, ShikiJS textmate-grammars-themes, Joplin, Visipedia.
Collaboration and community: WNB.rb, Rocky Enterprise Linux Foundation, Liqo, Apache Foundation, libdns, Skeleton UI and ggplot2.

Professional Experience

Systems Analyst (DevSecOps) @ Bates College *(2024–Present)*

Build and maintain enterprise systems supporting staff, faculty and the technical ILS team. 24/7 on-call for CVE triage and mitigation; help lead IaC and dev-CI parity college-wide.

- Built a property-based orchestrator and instrumentation harness for 1980s-era degree-management C in `Haskell` + `Python` (QuickCheck PBT, Cabal, FPM-packaged, podman-compose dev parity, GitLab AutoDevOps); turned an unautomatable legacy codebase into a verifiable, traceable, k8s-friendly workload.
- Overhauled and automated the event-management lifecycle (`C#`, `Go`, `Ansible`); helped lead horizontally-scalable `Apache Solr` adoption across multiple public/private indexing applications.
- Built internal ACME-first certificate-management and DNS libraries; helped lead SAML / Shibboleth / LTI integrations; contributed to enterprise secret management with `KeePassXC` inside a declarative Ansible workflow alongside firewalld, fail2ban and OTEL / LGTM / Tempo telemetry.

Computer Vision Software Engineer @ Macaulay Library, Cornell Lab of Ornithology *(2019–2021)*

Developed & launched **Merlin Sound ID**, a production fine-grained audio classification system used by millions worldwide. Stack: Objective-C + Swift (iOS), Python/TensorFlow (training). Led R&D on internal ML annotation tooling, model-evaluation web APIs, and end-to-end MLOps pipelines (TypeScript, React, Vue, Docker, BitBucket CI); built real-time inference demos in React Native and Swift; contributed to novel community-science fine-grained classification research with Visipedia.

Fabrication Laboratory Manager @ Cornell CALS Landscape Architecture (DLA Makerspace) *(2021–2022)*

Developed and taught rapid-fabrication curricula. Built C++ tiler / slicer tools with my teaching assistants; ran the team's GitHub & Linear project tracking.

Engineering & Contracting *(2017–Present)*

Operating as xoxd.ai as of 2024.

- **Web GIS, GDAL/OGR and R/Shiny**: National Park Service, Foundation for Healthy Communities, GPRED, NBRC, UNH and NH municipalities; community mapping, Photovoice and avian field tools; presented at AAG 2019.
- **Engineering**: Adaptive Motorsport control hardware and USB-HID software; contract bootloaders; hydraulic sawmill equipment R&D; business systems for small firms.
- **Infrastructure and product work**: build platforms, infrastructure automation, agent orchestration, authentication, scheduling and data integrations; independent ML work with MushroomObserver and Visipedia.

Client portfolio and public full stack works delivered available on request!

Community, Publications & Talks

Community:

- Ongoing engagement with the **Rocky Enterprise Linux Foundation**: Rocky 10 platform, real-time (PREEMPT_RT) kernel lanes and downstream CVE backport practice.
- *First Fellow*, D&M Makerspace, Plymouth State University (2017–2020); taught *Advanced GIS Programming*.
- Coordinated the regional COVID-19 PPE manufacturing response with the NH hospital system.
- *Membership Chair & 3D Printing Captain*, Ithaca Generator (2020–2022).
- *Founder*, Great Falls Tool Bus, a member-run Lewiston tool library (2026–). Built its public site and member-management infrastructure.

Education: **B.S., Interdisciplinary Computer Science**, Plymouth State University (2016–2019), *magna cum laude*; focus: *Ecological Education & Analytics*; GIS certification. Taught *Advanced GIS Programming* and *Intro to Electromechanics*.

Upcoming talk: XO Ruby, Montréal, October 17, 2026: iteration and inventiveness in machine learning.

Publications:

Sullivan, J. (2026). *Recovering Write-Protected NVMe SSDs Through USB-Bridge XRAM Injection: Bypassing the ASMedia ASM2362 Firmware Opcode Whitelist.* transscendsurvival.org/papers/recovery-paper.pdf.

Reitsma, L.R., Burns, C. & **Sullivan, J.** (2019). *Poecile atricapillus* (Black-capped Chickadee) Feeding *Catharus guttatus* (Hermit Thrush) Nestlings. *Northeastern Naturalist*, 26(2). doi:10.1656/045.026.0213

Sullivan, J. (2019). *Web GIS: Telling Stories & Solving Problems. American Association of Geographers (AAG) Annual Meeting*, Washington, DC.

Ten or more professional references (academic, industry and FOSS) available on request.